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Thermal properties of carbon fibers at very high temperature

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ABSTRACT

Thermal properties such as specific heat C_p, thermal diffusivity a, and thermal conductivity λ of carbon fibers are important parameters in the behaviour of the carbon/carbon composites. In this study, the specific heat and the thermal diffusivity are measured at very high temperatures (up to 2500 K). The experimental thermal conductivity estimated by the indirect relation $\lambda = a\rho C_p$ is presented as a function of the temperature. Validations are carried out on isotropic metallic (tungsten) and ceramic (Al_2O_3) fibers. Measurements were obtained on three carbon fibers (rayon-based, PAN-based and pitch-based). Thermal conductivity results allow us to classify fibers from the most insulated to most conductive. The main result is that insulated carbon fibers have an increasing thermal conductivity when the temperature and the heat treatment temperature rise. Relationships between thermal conductivity and the structural properties (L_c and d_{002}) of such carbon fibers are studied. We also describe the influence of heat treatment on the thermal conductivity of carbon fibers.

1. Introduction

At extremely high temperatures, carbon/carbon composite materials exhibit a high Young's modulus, thermal conductivity and thermal expansion [1,2]. These properties are particularly attractive for aeronautics (braking systems), aerospace (heat shields) and military applications – although global macroscopic properties are generally sufficient to design and model the behaviour of those materials. Particular uses, at an extremely high temperature and with a high thermal gradient, we need to investigate the properties of the constitutive parts (fibers) of the materials. The literature gives little data on the properties of carbon fibers and matrices at high temperatures [3,4]. This study focuses on thermal properties measurements at very high temperatures, it is included in a large program devoted to mechanical [5], thermal [6,7] and thermomechanical [8] characterisation of carbon fibers.

The difficulties of such measurements are due to the micro scale of the fibers (≈10 μm) and the wide range of temperatures (300–2700 K). An experimental device and a data processing method were developed for the measurements of the specific heat [6] and the thermal diffusivity [7]. They allow characterisation with extreme accuracy, reducing measurements uncertainties to less than ±5% for both the specific heat and the thermal diffusivity.

The first aims of this study are the measurements of the specific heat and the longitudinal thermal diffusivity on various kinds of fibers at very high temperatures. Then, we will use the calculation of the thermal conductivity (Eq. (1)) with the measured specific heat (C_p), thermal diffusivity (a) and mass density (ρ) to understand the relationships between the measured values of the thermal conductivity and the structural properties of the fibers.

$$\lambda(T) = a(T)\rho(T)C_p(T) \tag{1}$$

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Finally, we will set out classification of the thermal conductivity of the fibers in order to establish that such measurements at very high temperatures improve the understanding of the thermal behaviour and the structural properties of such carbon fibers.

2. Measurement method of the thermal properties

2.1. *Specific heat*

Two experimental apparatus are used for the measurements of the specific heat, the differential scanning calorimetry (DSC) up to 1900 K and the drop calorimetry, for higher temperatures up to 3000 K. The main drawback of these techniques is the minimum sample mass (100 mg) required. Even if some authors have developed the AC calorimetric method [9,10], measurements on single carbon fibers have never been performed. In previous works [7], we developed a method for the measurements of the specific heat on single carbon fibers at very high temperatures. The principle consists in heating-up the fiber using the Joule effect. The electrical tension applied on the boundaries of the fiber is modulated at different frequencies. The temperature response, also modulated, is measured by an infrared sensor over a large region of the isothermal sample. Then, the phase shift between the electrical excitation and the thermal response is recorded at different frequencies. Finally, an analytical model allows an estimation of the specific heat value at the average temperature of the experiment.

2.2. *Thermal diffusivity*

Several kinds of experimental techniques have been developed to perform the measurement of the thermal diffusivity of single filaments. At very low temperatures (5 K to 300 K), the contact method is often used [11]. The thermal conductivity is simply calculated using Fourier's law, taking into account the heat losses along the sample. A similar method is proposed [12] for the measurements at room temperature. At high temperatures, from 300 K to 800 K, periodic excitation from a halogen lamp and contact detection by thermocouple has also been developed [13] on a commercial apparatus (Sinku-Riko Inc., model PIT-1) described herein [14]. The Mirage method, achieved for thermal characterisation [15], is very difficult to apply to a single fiber. The detection is achieved by measuring the deflection of a laser beam assumed to be as linearly dependent on the temperature. The photoreflective experiment uses the reflection of a second laser beam focalised close to the excitation to measure a signal proportional to the temperature response of the sample. This method has been developed for the thermal characterisation of fibers [16]. Recently, the thermal diffusivity of the fibers has been measured at room temperature with a photothermal microscope using the phase shift between the irradiating laser beam and the detected emission [17].

In previous works [6], we developed a method for the measurement of the thermal diffusivity of single carbon fibers at very high temperatures. In comparison with the literature, the extremely high temperatures (700–2500 K) and the data processing method are the main innovations of the work. The principle of the method consists in heating-up the fiber by Joule effect. A periodic thermal excitation is achieved with a laser beam focalised on the surface of the sample. The thermal temperature response is obtained through an optical (infrared) sensor. Then an analytical thermal model is used for the estimation of the thermal diffusivity.

3. Description of the experimental apparatus

The experimental device (Fig. 1), was developed on a bench designed for mechanical, thermal and thermomechanical characterisation [18]. The fiber, glued at each end on two car-

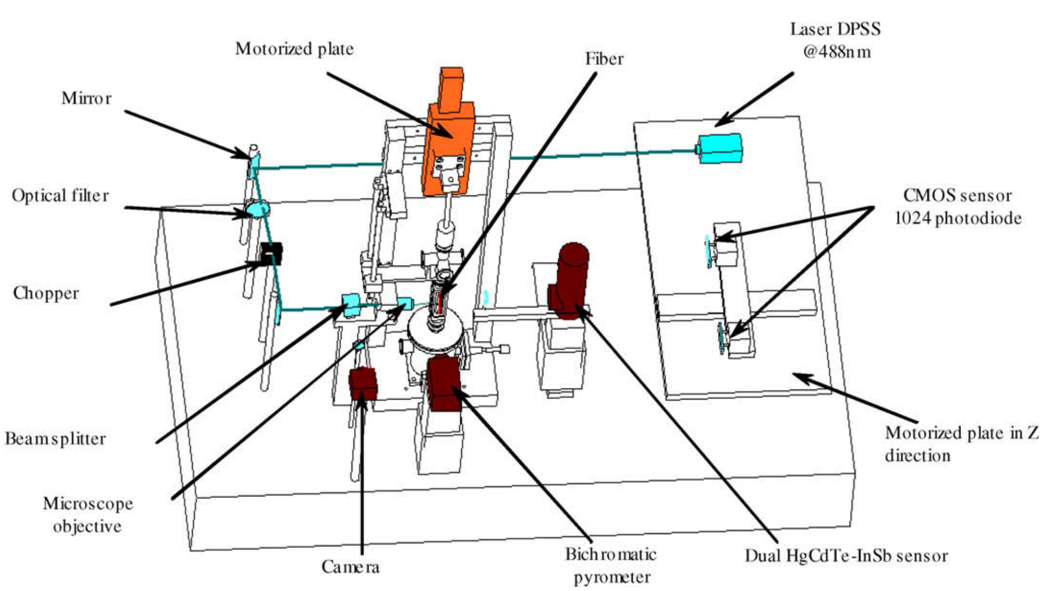

Fig. 1 – Scheme of experimental device.

bon grips, is heated up by Joule effect. The temperature of the fiber is quasi isotherm ($T_{center} - T_{surface} < 2$ °C, [7]). A very low pressure ($P < 10^{-3}$ Pa) is maintained around the fiber in a silica glass enclosure. A bichromatic pyrometer [0.75–1.05 μm] measures the absolute temperature of the fiber over a zone of detection zone of 300 μm diameter. An infrared detector (PbSn, 1–3 μm, 1 mm^2 sensitive area, liquid N_2 cooled) combined with two plano-convex $BaCF_2$ lenses (25 mm and 50 mm focus length) is used for the temperature response measurement. The position of the fiber is adjusted with a micrometer-moving system located inside the enclosure and remote-controlled.

For the measurements of the specific heat, the energy modulation of the fiber is obtained thanks to a stabilised electrical power supply coupled with a lock-in amplifier, which measures the phase shift between the electrical excitation and the temperature response. The phase shift measurements are transferred to a computer where the numerical processing is performed.

For the measurements of the longitudinal thermal diffusivity, the thermal excitation is imposed by a modulated multimode Argon laser beam (power up to 200 mW, frequency from 0.1 to 50 Hz). The beam is focalised with a microscope objective (magnitude ×50, working distance 16 mm) on an irradiated zone of ≈2 μm diameter. This spot is also experimentally monitored during the experiment with a CCD camera. The infrared detector is used for the measurement of the temperature response, along the fiber (with motorised displacement), over a detection zone of 125 μm diameter. This spot, equivalent to a detection length L along the fiber, is evaluated from geometrical optics formulas, and systematically measured at the beginning of each experiment using a lateral scanning of the fiber by the detector.

During the experiment, a lock-in amplifier determines the magnitude and the phase shift between the electromechanical chopper and the temperature response detected by the infrared sensor. These measurements for several frequencies and positions of the infrared detector are transferred to a computer. Then, data processing routine yields the slopes of the linear parts automatically detected on the records. This operation is obtained from a local derivation of the experimental curves, following by a numerical filter.

4. Structural properties of the fiber specimens

4.1. *Description of the carbon fibers*

In this study, three kinds of carbon fibers were measured: TC2, PANEX 33 and P100. PANEX 33 fiber (ZOLTEK Corp.) elaborated at 1600 K has a PAN precursor. The PANEX 33 was heat-treated at two temperatures (1900 and 2500 K). Finally, the P100 fiber (AMOCO Performance Products Co., formerly Union Carbide Co.) comes from an anisotropic pitch precursor with heat treatment above 2700 K by the supplier and the TC2 fiber (from SAFRAN) comes from a rayon precursor. The TC2 fiber elaborated at 1500 K was heat-treated at 2500 K. For each fiber, the highest measured temperature never goes over HTT. Therefore, we assumed that fibers would be stable during the measurement process. The characteristics summarized in Table 1 indicate that a wide spectrum of fibers is covered. Except for the P100 fiber, these characteristics were measured in our laboratory.

4.2. *Structural characterisation of the fibers*

L_c, L_a and d_{002} correspond to standard parameters used in the description of carbon structures. L_a is the longitudinal dimension of crystallites; L_c is the lateral dimension and d_{002} is the mean distance between two successive layers. These parameters have been determined using X-ray diffraction [19]. Here, L_c and d_{002} were determined using the Scherrer equation from the positions of the diffraction maxima and the width at half-maximum intensity of the (002) and (110) peaks. The orientation distribution of graphitic planes was determined by X-ray analysis of the fibers, as described by Ruland [20]. It was obtained from an azimuthal spread of the 002-reflection arc. This provides an indication of the preferred orientation of the basic structural units (BSU) relative to the fiber axis. The orientation parameter Z (°) is the full width at half the maximum intensity obtained in the azimuthal scan. Fiber texture was described using both the distribution of the intensity of scattering at angle Φ and the orientation parameter Z. Measured parameters describing the structure and the texture of fibers are presented in Table 1. We notice that each parameter obviously varies according to the stiffness of the fibers: L_c increases, d_{002} decreases, Z decreases, Young's modulus E increases and mass density ρ increases.

Table 1 – Structural characteristics of fibers at room temperature and thermal conductivity measured at 1500 K.

Fiber name	HTT (K)	d_{002} (pm)	L_c (nm)	Z (°)	E (Gpa)	ρ[d] ($g\,cm^{-3}$)	λ[e] ($W\,m^{-1}\,K^{-1}$)
TC2	1500[a]	371	1	–	34[b]	2	15.4
	2500	356	1.4	–	33[b]	1.4	28.5
PANEX 33	1600[a]	349	1.9	24.5	230[b]	1.75	75
	1900	345	2.4	23	253[b]	1.77	82.6
	2500	343	4.2	19	297[b]	1.96	91.5
P100	2700[a]	338.8	10.6	6.4	690[c]	2.15	170.7

a As received.
b Measured by tensile tests [5].
c From supplier.
d Measured using a helium pycnometer.
e Measured in this study at 1500 K.

This trend indicates the presence of crystallites with an increasing size, separated by a shorter distance between two successive layers and with a more pronounced orientation in the fiber axis direction. This shows an increasing degree of anisotropy.

5. Results and discussion

5.1. *Validation on isotropic fibers*

Validation of the measurements of specific heat and longitudinal thermal diffusivity was performed using a metallic tungsten isotropic fiber. Measured values were compared with the well-known thermal properties of the bulk [21]. Another validation was made on isotropic ceramic fiber: Al_2O_3 Nextel 720 in comparison with bulk [22].

In Fig. 2, the measured thermal properties of the tungsten filament of 18 μm diameter and 50 mm length, are compared to the bulk properties [21]. For the specific heat Fig. 3a, the difference between measured values and bulk data is lower than 5%. For the longitudinal thermal diffusivity Fig. 3b, the difference with the literature is about 10%. Therefore, we assumed that a rather good agreement does exist. This allows the validation of the measurements methods and of the experimental device.

Then, measurements on the electrical insulator Al_2O_3 Nextel 720 isotropic ceramic fiber were performed. In order to heat up the fiber by Joule effect, a pyrocarbon deposit is first made by CVD. The obtained thickness is below 100 nm. With this coating, the N720 fiber was heated up by Joule effect and the measurements of the thermal properties were performed (Fig. 3). The heating ramp applied to this fiber of 10 μm diameter and 50 mm long, was from 900 K to 1500 K by steps of 100 K. The results Fig. 3a show that the specific heat presents values close to data given in the literature [22] (difference lower than 3%). The calculated (Eq. (1)) thermal conductivity Fig. 3c with the measured specific heat (Fig. 3a) and the thermal diffusivity (Fig. 3b) are also in rather good agreement with the bulk data [22]. The deviations of the measurements are about 10% at any temperature. Furthermore, we can observe that the thin pyrocarbon coating has no effect on the evaluation of the thermal properties at a high temperature.

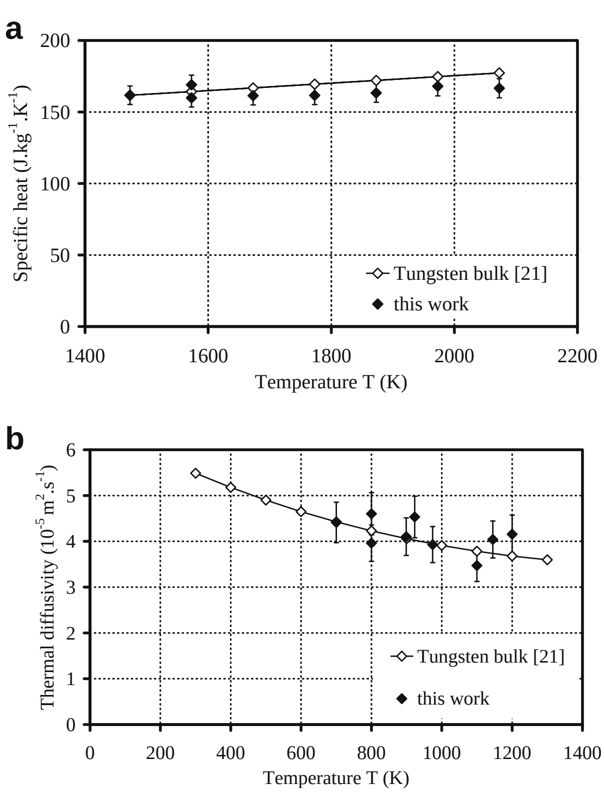

Fig. 2 – Measurements on tungsten metallic isotropic filament: (a) specific heat, (b) thermal diffusivity.

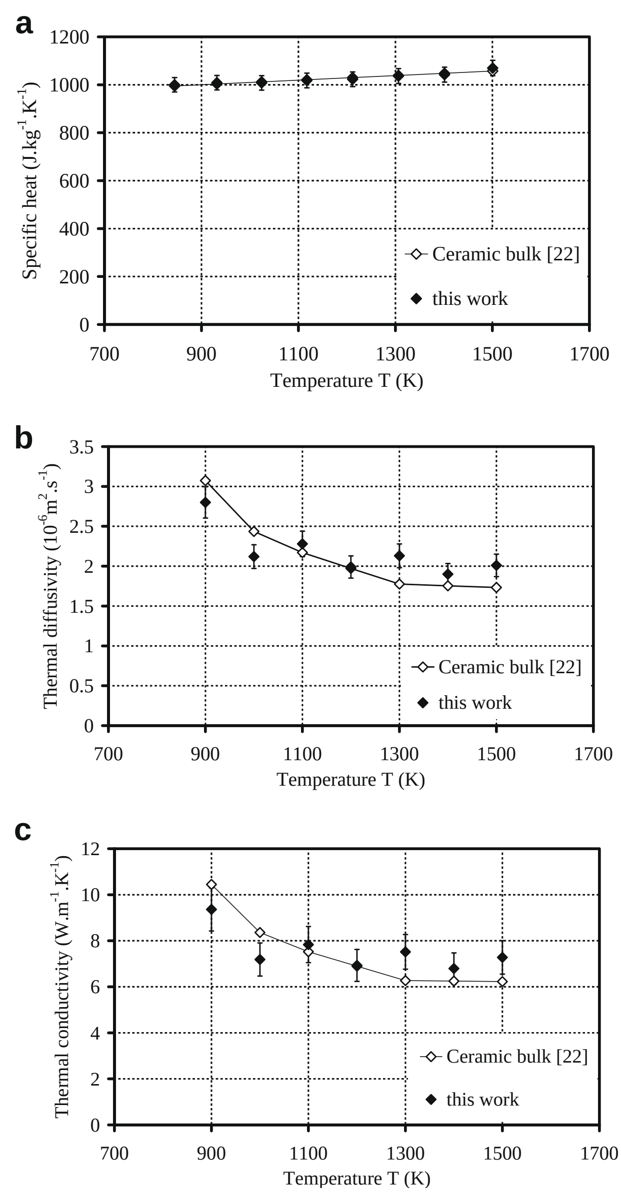

Fig. 3 – Measurements on Nextel 720 ceramic isotropic filament: (a) specific heat, (b) thermal diffusivity and (c) calculated thermal conductivity.

5.2. Measurements on carbon fibers

Previous results obtained on conductive metallic and insulated ceramic fibers allow us to perform measurements and advanced material interpretation on various carbon fibers. Measurements were performed on the three carbon fibers described earlier (see Table 1). The following experimental measurements are presented in ascending rank from most isotropic to most anisotropic fibers.

The measurements of the thermal properties on raw and HTT of 2500 K TC2 fibers are plotted (Fig. 4) as a function of the temperature from 500 K to 2500 K. The specific heat, Fig. 4a, is compared to the graphite bulk data [23]. For raw and heat-treated fibers, the measured values are quite close from those in the bulk. On this rayon fiber, we observed that the heat treatment decreases the specific heat at any temperature. Moreover, for a heat-treated fiber a very important increase of the specific heat starts at 2100 K. This could be explained by the defect damping due to the rise of the temperature level. In Fig. 4b, the experimental data obtained on the thermal diffusivity show that the thermal diffusivity increases linearly with the temperature. In contrast with the specific heat, the heat treatment doubles treated the fiber's thermal diffusivity.

Measurements of thermal properties on raw and HTT up to 1900 K PANEX 33 fibers are plotted (Fig. 5) as a function of the temperature from 500 K to 2500 K. Fig. 5a shows the effects of the thermal treatments on the specific heat. This is consistent

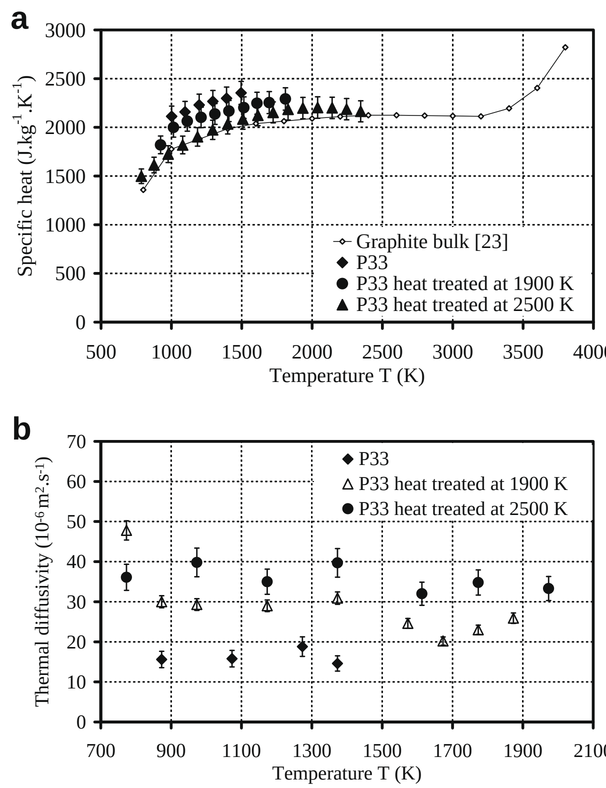

Fig. 5 – Measurements on PAN-based PANEX 33 carbon fibers: (a) specific heat; (b) thermal diffusivity.

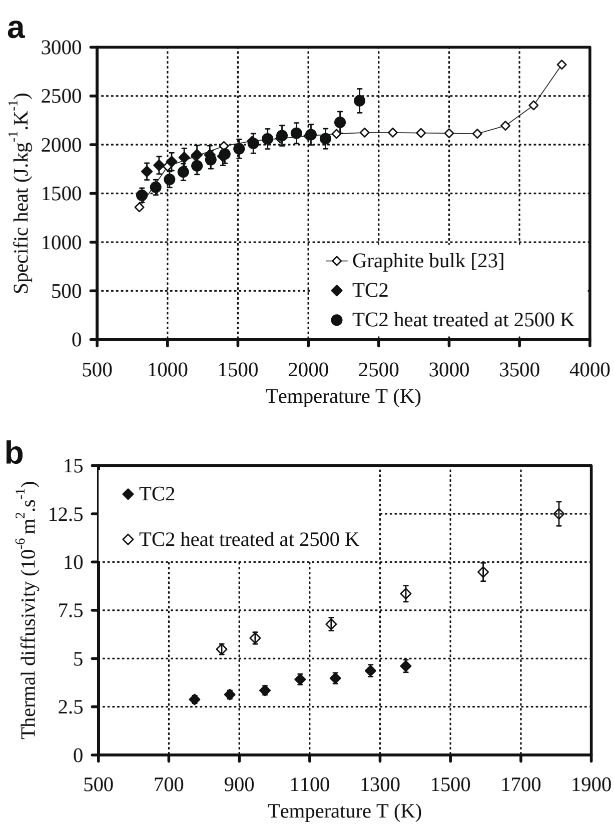

Fig. 4 – Measurements on rayon-based TC2 carbon fibers: (a) specific heat; (b) thermal diffusivity.

with the structural properties results (Table 1). For the raw PANEX 33 fiber, the specific heat is important compared to that of the graphite single crystal. When the HTT increases from raw to 2500 K, the specific heat decreases. For the PANEX 33 heat-treated at 2500 K, measured values are quite close to graphite's. This is due to the textural reorganisation of this turbostratic fiber. The behaviour of the PANEX 33 fiber thermal diffusivity is plotted Fig. 5b. For a raw fiber, the evolution of the thermal diffusivity is equivalent to the raw TC2 fiber. When the HTT increases from raw to 2500 K, the thermal diffusivity increases. For a HTT of 1900 K and 2500 K, the thermal diffusivity is quasi constant with a light decrease versus the temperature increase.

In Fig. 6, thermal properties on P100 fiber are plotted as a function of the temperature from 500 K to 2500 K. Fig. 6a shows that this fiber with a graphite-like structure has a specific heat higher than graphite bulk. We assumed that this difference was due to the measurement processes. In fact, this fiber of high electrical conductivity is difficult to characterise with the apparatus developed [7]. Here measurement uncertainties are of about 20% due to the technical problems on the measurements of the electrical resistivity. In Fig 6b, the thermal diffusivity has been successfully measured with an uncertainty of less than 5%. This fiber, with highly ordered graphite domains, has a thermal diffusivity that decreases with the temperature. This behaviour is similar for PANEX 33 fibers with HTT up to 1900 K.

In order to rank the studied carbon fibers from the most insulated to the most conductive, the thermal conductivity λ

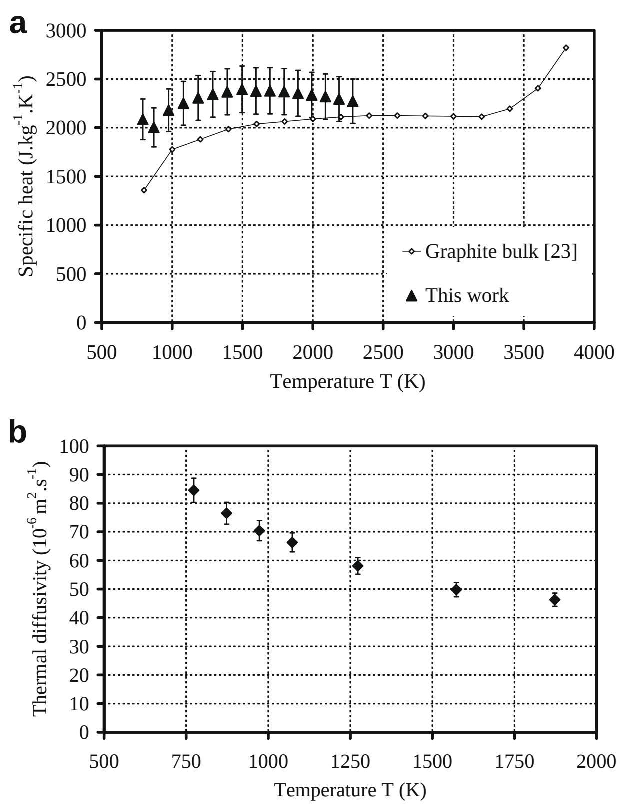

Fig. 6 – Measurements on pitch-based P100 carbon fiber: (a) specific heat; (b) thermal diffusivity.

calculated from (Eq. (1)) is plotted (Fig. 7) as the function of temperature for each carbon fibers. The results obtained at that point allow us to rank fibers as follows: TC2 < PANEX 33 < P100. Further, turbostratic fibers (as TC2 and raw PANEX 33) show a quasi-linear increasing of thermal conductivity versus analysis temperature. In contrast, ordered graphite fibers, like PANEX 33 treated up to 1900 K present an increasing thermal conductivity up to the temperatures related to the fiber organisation. Then, the thermal conductivity decreases until an asymptotic state is reached. In comparison, for P100 fiber, the thermal conductivity is higher and always decreases due to the well-known umklapp phenomena. This very interesting behaviour shows that low conductive carbon fibers have a thermal conductivity that increases with the temperature. This is possibly a consequence of the defect damping with HTT and increasing temperature.

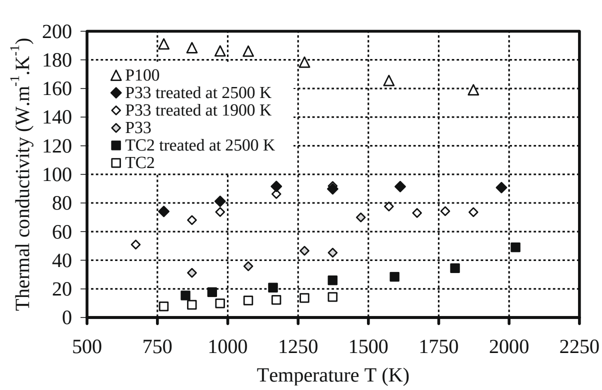

Fig. 7 – Thermal conductivity measurements of rayon-based, PAN-based and pitch-based carbon fibers.

5.3. *Thermal behaviour with structural and mechanical properties*

In order to investigate relationships between the thermal behaviour of carbon fibers and the structural properties, we compare the thermal conductivity measured at 1500 K with the others properties in Table 1 obtained at room temperature. To evaluate all the fibers, we chose the temperature equal to 1500 K due to the maximum measured temperature of the raw TC2 fiber. As shown at ambient temperature [24,25], the thermal conductivity decreases as the interlayer spacing d_{002} increases and increases as the crystalline size L_c increases. These results establish that the fiber's degree of ordering of crystalline and heat treatment are responsible for the change of the thermal conductivity values of such carbon fibers. The dependence of the thermal conductivity with the Young's modulus is achieved. As assumed [26], the thermal conductivity increases with Young's modulus increases. This behaviour is the result of the direct effects of the structural crystallite parameters on mechanical, thermomechanical and thermal properties.

6. Conclusion

The experimental device and the data processing method previously developed are useful for measuring the thermal properties of fibers at very high temperatures. The thermal properties of metallic, ceramic and carbon fibers (rayon-based, PAN-based and pitch-based) have been measured in this study. These kinds of results allow data acquisition in order to design and model the behaviour of composite materials at very high temperatures.

With the data obtained on the thermal conductivity, it is possible to rank fibers from the most insulated to the most conductive (TC2 < PANEX 33 < P100). We assumed that rayon-based fibers are more insulated than PAN-based fiber. This latter one is also less conductive than pitch-based carbon fibers. For rayon-based and raw PAN-based fibers, the thermal conductivity increases with the temperature. For heat-treated PAN-based fibers, an increase then a decrease of thermal conductivity is observed. Moreover, for pitch-based fiber, the thermal conductivity always decreases. The major result comes from the behaviour of low conductive carbon fibers that have an increasing thermal conductivity when the HTT and the temperature rise.

In addition, such measurements also involved the effects of the structural properties of the carbon fibers on the thermal behaviour. As measured by X-Ray analysis, the most isotropic fiber (TC2) has the lowest thermal conductivity whereas the most anisotropic fiber (P100) has the highest thermal conductivity. The influence of the HTT is important too (see PANEX33 and TC2). We have observed that an increase of HTT on PAN-based and rayon-based fibers had some effects on the thermal behaviour. Then, the specific heat decreases whereas the thermal diffusivity and the thermal conductivity increase with increasing HTT.

Finally, relationships between the thermal conductivity and the crystallite size allow us to demonstrate the link between the thermal behaviour and the structural properties. This result confirms that the textural organisation of the carbon fibers controls the thermal conduction of carbon fiber materials.

Acknowledgements

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